Names Vs Qualifications:
A Simulated Audit Study of the Intersection of Name-Ethnicity and Gender on CV Callback Rates in UK Corporate Hiring

---

LIS MASc | 2025/26 | FT Trials & Errors | 2500014873

AI Declaration: Claude's Artificial Intelligence model Opus 4.6 was adopted to create the simulated dataset used for analysis in this study as well as to create the code cell titled '#10. write outputs' to speed up the process.

Word Count: 2065

GitHub Repo: https://github.com/tulin-b/trials_and_errors-assessment2.git

---

Introduction

Labour-market discrimination remains among the most thoroughly documented forms of social inequality in industrialised economies. In a comprehensive synthesis, Quillian, Pager, Hexel and Midtbøen (2017) pooled 28 correspondence-style hiring experiments conducted in the United States between 1989 and 2015 and concluded that White applicants received 36% more callbacks, on average, than equivalently qualified African-American applicants, with no detectable improvement across the 26 year window. Correspondence experiments are the methodological gold standard for isolating discriminatory behaviour (Pager, 2007) as, by submitting matched fictitious applications that vary only in a randomised demographic signal, they identify the causal contribution of perceived identity while holding qualifications, employment history and self-presentation constant (Bertrand & Mullainathan, 2004).

Parallel European literature shows the structural pattern holds despite cross-national variation in targets. Kaas and Manger (2012) submitted matched applications bearing typical German or Turkish surnames to identical vacancies where Turkish-named applicants received approximately 14% fewer callbacks on average and 24% fewer at small firms, with the gap attenuating when applications included diagnostic reference letters. This provides evidence consistent with statistical discrimination under information asymmetry. The GEMM cross-national field experiment (Koopmans, Veit & Yemane, 2019) sent 19,181 fictitious applications across five European countries and reported the largest callback penalties for Turkish-origin and other culturally-distant applicants.

These main-effect estimates are rarely stratified by gender even though feminist scholarship has argued that disadvantages rarely combine additively. Crenshaw's (1989) critique of antidiscrimination doctrine demonstrated that Black women experienced harms legally invisible to both race and sex jurisprudence because they occurred at the intersection of the two. Subsequent evidence has been mixed, with Pedulla (2014) documenting reversals in which certain intersections benefited from offsetting stereotypes, whereas Bursell (2014) reported that Middle-Eastern women in Sweden

suffered the largest penalties. Whether these effects compound or offset in UK corporate hiring remains unclear. The UK's Turkish-origin population is smaller than Germany's, and its sectors like law, consulting, and similar professional services are characterised by prestige-based assortative hiring (Ashley & Empson, 2013; Rivera, 2012), both of which may alter the intersectional dynamic.

The present study adopts an explicitly falsificationist stance (Popper, 1959/2002). Rather than seeking to prove that discrimination exists, it pre-specifies the patterns of data that would reject each hypothesis, and treats survival of a sincere refutation attempt as tentative corroboration. Because the data are simulated from parameters fixed a priori, the design functions as its own pre-analysis plan (Munafò et al., 2017).

Four hypotheses are tested:

> ***H1a.*** *CVs bearing Turkish-origin names are less likely to receive a callback than otherwise-identical CVs bearing English-origin names in a simulated UK corporate hiring context.*
>
> ***H1b.*** *CVs bearing female-signalled names are less likely to receive a callback than male-signalled names.*
>
> ***H1c.*** *The Turkish-origin penalty is larger for female-signalled than for male-signalled CVs (intersectional interaction).*
>
> ***H0.*** *Callback rates do not vary by name-ethnicity, gender or their interaction.*

Methods

Design

A 2 (name-ethnicity: Turkish-origin, English-origin) × 2 (gender: female-signalled, male-signalled) fully between-CV factorial design was employed. Callback was coded by ‘1’ if the employer invited the applicant to interview and ‘0’ if otherwise, serving as the binary dependent variable. The design follows the correspondence-test template established by Bertrand and Mullainathan (2004) considering every CV was held constant in degree class, employment history, referees and formatting, varying only in the applicant's forename–surname pair.

Participants and target sample

The data analysed are simulated and no real human or organisational participants were involved. A field version would target UK employers advertising entry-level administrative or consulting roles on public job boards. A stratified sample of 400 vacancies (two CVs each; N = 800 CVs) would be drawn to match the statistical power of Kaas and Manger (2012) and Pager (2007).

Materials and operationalisation

Name-ethnicity is operationalised through forename–surname pairs matched on frequency and religious connotation. English-origin names drawn from the UK Office for National Statistics baby-name register (e.g., Emma Smith, James Brown) and Turkish-origin names drawn from the Türkiye Nüfus ve Vatandaşlık İşleri Genel Müdürlüğü (Turkish General Directorate of Population and Citizenship Affairs) register (e.g., Ayşe Yılmaz, Mehmet Demir). Gender is signalled by forename following the convention in the correspondence-study literature (Kaas & Manger, 2012; Koopmans et

al., 2019). All other CV features including degree classification, institution prestige tier, referees, employment tenure are held constant across conditions.

Data generation

Simulated callback outcomes were drawn from a Bernoulli distribution with $\text{logit}(p) = \beta_0 + \beta_1 \cdot \text{Turkish} + \beta_2 \cdot \text{Female} + \beta_3 \cdot (\text{Turkish} \times \text{Female})$; $\beta_0 = \log(0.25 / 0.75)$ yielded a 25% baseline callback rate for English-male CVs, and odds ratios of 0.55 for Turkish, 0.95 for female and 0.75 for the interaction approximate the effect sizes reported by Kaas and Manger (2012) and Pedulla (2014). The random seed was fixed at 42 with the full Python code accessible in the provided GitHub Repo.

Analysis plan

Descriptive callback rates and 95% Wilson confidence intervals (Newcombe, 1998) were computed per cell. A Pearson χ^2 test on the ethnicity × callback contingency table quantified the simple bivariate association. The pre-registered inferential test was a binary logistic regression of callback on name-ethnicity, gender and their interaction, fitted by maximum likelihood where coefficients were exponentiated to odds ratios with 95% Wald intervals. Alpha was fixed at .05 two-sided and the decisive test of H1c was the interaction coefficient. A supplementary Bayesian analysis used uniform Beta (1, 1) priors on each cell's callback probability and quantified the posterior probability that Turkish-origin CVs were less likely to be called back than English-origin CVs.

Ethical considerations

Since the data are simulated, no individuals or employers were deceived and no ethical review was required. The design, however, maps directly onto a real-data version that would require significant scrutiny, and therefore ethics form part of the methodological justification. Applying Emanuel, Wendler and Grady's (2000) seven criteria, a field version would face once irreducible tension: informed consent is structurally incompatible with audit deception and only partially mitigable through a post-hoc debrief. Independent review would also need to span both a university ethics committee and legal counsel, given that correspondence studies occupy a grey area under the Equality Act 2010. This deception is often contrasted with the ethical failings of Milgram (1963), yet differs in a morally relevant respect as it targets organisations rather than named individuals and carries no foreseeable psychological or physical harm.

Results

Descriptive callback rates by condition are displayed in Figure 1. CVs bearing English-origin male names received a callback rate of 24.5% (49/200), English-origin female names 26.0% (52/200), Turkish-origin male names 14.0% (28/200), and Turkish-origin female names 11.5% (23/200). The Pearson $\chi^2(1) = 20.31$, $p < .001$ shows Turkish-named applicants were ~49% as likely to receive a callback as English-named.

The pre-registered logistic regression (Table 1) confirmed a main effect of name-ethnicity. Applicants with Turkish-origin names had odds of callback 0.50 times those with English-origin names, 95% CI [0.30, 0.84], $p = .008$. Neither the main effect of gender (OR = 1.08, 95% CI [0.69, 1.70], $p = .730$) nor the Turkish × Female interaction (OR = 0.74, 95% CI [0.35, 1.55], $p = .421$) reached statistical significance, although the interaction point estimate was directionally consistent with intersectional

compounding. The omnibus likelihood-ratio test was significant ($p < .001$) with McFadden pseudo-$R^2 = .027$, reflecting the dominance of the ethnicity main effect.

The Bayesian supplementary analysis, based on uniform Beta(1, 1) priors and 50,000 posterior draws per cell, gave a posterior probability that the Turkish-name callback rate was lower than the English-name rate of greater than .9999, converging on the frequentist conclusion with high confidence.

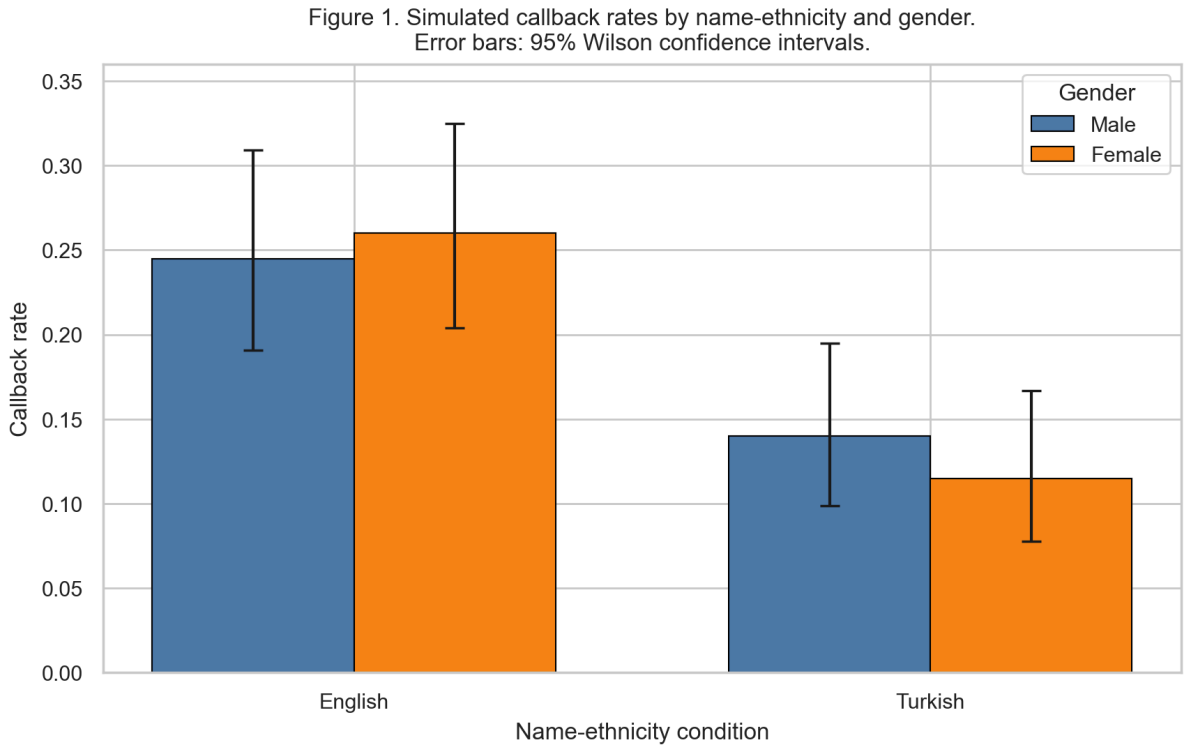

Figure 1. Simulated callback rates by name-ethnicity and gender (N = 800). Error bars represent 95% Wilson score confidence intervals. Callback rates are 24.5% (English-Male), 26.0% (English-Female), 14.0% (Turkish-Male) and 11.5% (Turkish-Female).

Predictor	OR	95% CI (low)	95% CI (high)	p
Intercept (English, Male)	0.33	0.24	0.45	< .001
Turkish-origin name	0.50	0.30	0.84	.008
Female-signalled name	1.08	0.69	1.70	.730
Turkish x Female interaction	0.74	0.35	1.55	.421

Table 1. Logistic regression of callback on name-ethnicity, gender and their interaction (N = 800). Odds ratios are reported with 95% Wald confidence intervals. Reference categories are English-origin and male-signalled.

In terms of the pre-specified hypotheses, H1a was supported as there was clear evidence against H0 with respect to name-ethnicity. H1b was not supported as the data were consistent with H0 with respect to gender. H1c was not supported as the data did not provide evidence against H0 with respect to the interaction, although the direction of the estimate did not contradict it.

These findings replicate, in a simulated UK setting, the central observation of the European correspondence-study literature (Kaas & Manger, 2012; Koopmans et al., 2019). They diverge from Bursell's (2014) Swedish evidence of intersectional compounding for Middle-Eastern women, though that comparison is imperfect as Bursell's category pooled Turkish, Arabic and Persian names, and the Swedish and UK labour contexts differ in ways that may shape the intersectional pattern.

Discussion

This study tested whether name-ethnicity, gender and their interaction predicted CV callback outcomes in a simulated UK corporate hiring context. Three findings structure this interpretation. First, Turkish-origin names were associated with reduced odds of callback (OR = 0.50), a magnitude broadly comparable to the 50% gap reported by Bertrand and Mullainathan (2004) for African-American names in the United States and to the fourteen-percentage-point gap of Kaas and Manger (2012) in Germany. Second, there was no detectable main effect of gender. Third, contrary to the intersectional compounding hypothesis, the Turkish × Female interaction did not reach significance.

The null intersectional result is scientifically informative. Under the falsificationist account (Popper, 1959/2002), a hypothesis gains credibility when it survives a sincere attempt to refute it and forfeits credibility when it does not and at the observed sample size, H1c did not survive its first test. Three non-exclusive interpretations are available. The effect may be real but smaller than a single N = 800 study can reliably detect so the 95% interval for the interaction odds-ratio, [0.35, 1.55] would be expected to shrink with a larger sample. Alternatively, intersectional compounding may be moderated by industry or by religious signalling beyond the name alone (Bursell, 2014), neither of which was varied here. Finally, the hypothesised multiplicative structure may itself be misspecified: gender and ethnicity penalties may operate on different scales, so that attempting to detect their interaction within a single logistic model could be the wrong question.

The robust main effect of name-ethnicity invites discussion of mechanism. Two competing accounts dominate labour economics. Taste-based discrimination (Becker, 1957) attributes the gap to employer or colleague preferences against out-group applicants, independent of productivity. Statistical discrimination (Arrow, 1973; Phelps, 1972) attributes it instead to rational use of group membership as a proxy for unobserved productivity characteristics. Kaas and Manger (2012) found the gap narrowed with additional diagnostic information, consistent with statistical discrimination; Koopmans et al. (2019) found an opposite pattern, consistent with taste-based rather than statistical discrimination, for Turkish and culturally-distant groups in Germany. The present design cannot adjudicate between these accounts but could easily be extended to do so by varying whether CVs

include extra signals of applicant quality (for example, reference letters), following Kaas and Manger (2012).

Several limitations must be acknowledged. The simulation approach imposes effect sizes rather than recovering them and its contribution is methodological. Name-based operationalisation of ethnicity is reductive and conflates perceived religion, class and national origin. The binary gender coding ignores non-binary and trans applicants. Callback is a coarse proxy for hiring outcomes and cannot capture downstream gaps in wages or promotion. More broadly, reducing complex constructs (racism, misogyny, intersectional disadvantage) to a single binary indicator risks what Porter (1995) terms 'trust in numbers' at the expense of the narrative and ethnographic evidence that gives those numbers their meaning.

A final limitation deserves explicit acknowledgement. The data analysed here are simulated, not collected. Parameters were chosen from the published literature rather than estimated from observation, and the findings therefore illustrate the behaviour of a plausible statistical model rather than the behaviour of real UK employers. Conclusions about the UK hiring context are conditional on those imposed parameters, and the study's contribution is best understood as methodological and pedagogical. Only a properly structured field experiment with ethical approval, legal scrutiny and independent review would warrant substantive empirical claims in this field.

Taken together, these results position name-ethnicity as a first-order filter in UK corporate screening, large enough to detect at modest sample sizes, whereas gender and its intersection with ethnicity appear to be second-order effects requiring larger studies to characterise. If this ordering is confirmed by a pre-registered field replication, the policy implication is direct: name-blind screening should yield larger observable gains at shortlisting than gender-targeted interventions alone. Three extensions would test the claim: a pre-registered field version compliant with the Equality Act 2010; a factorial extension crossing ethnicity × gender × religious signal to disentangle Turkish national origin from perceived Muslim identity; and a blind-review intervention modelled on Goldin and Rouse (2000), converting a correlational pattern into a causal policy test.

Bibliography

Arrow, K. J. (1973). The theory of discrimination. In O. Ashenfelter & A. Rees (Eds.), Discrimination in labor markets (pp. 3–33). Princeton University Press.

Ashley, L., & Empson, L. (2013). Differentiation and discrimination: Understanding social class and social exclusion in leading law firms. Human Relations, 66(2), 219–244. https://doi.org/10.1177/0018726712455833

Becker, G. S. (1957). The economics of discrimination. University of Chicago Press.

Bertrand, M., & Mullainathan, S. (2004). Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination. American Economic Review, 94(4), 991–1013. https://doi.org/10.1257/0002828042002561

Bursell, M. (2014). The multiple burdens of foreign-named men: Evidence from a field experiment on gendered ethnic hiring discrimination in Sweden. European Sociological Review, 30(3), 399–409. https://doi.org/10.1093/esr/jcu047

Crenshaw, K. (1989). Demarginalizing the intersection of race and sex: A Black feminist critique of antidiscrimination doctrine, feminist theory and antiracist politics. University of Chicago Legal Forum, 1989(1), 139–167.

Emanuel, E. J., Wendler, D., & Grady, C. (2000). What makes clinical research ethical? JAMA, 283(20), 2701–2711. https://doi.org/10.1001/jama.283.20.2701

Goldin, C., & Rouse, C. (2000). Orchestrating impartiality: The impact of “blind” auditions on female musicians. American Economic Review, 90(4), 715–741. https://doi.org/10.1257/aer.90.4.715

Kaas, L., & Manger, C. (2012). Ethnic discrimination in Germany's labour market: A field experiment. German Economic Review, 13(1), 1–20. https://doi.org/10.1111/j.1468-0475.2011.00538.x

Koopmans, R., Veit, S., & Yemane, R. (2019). Taste or statistics? A correspondence study of ethnic, racial and religious labour-market discrimination in Germany. Ethnic and Racial Studies, 42(16), 233–252. https://doi.org/10.1080/01419870.2019.1654114

Milgram, S. (1963). Behavioral study of obedience. The Journal of Abnormal and Social Psychology, 67(4), 371–378. https://doi.org/10.1037/h0040525

Munafò, M. R., Nosek, B. A., Bishop, D. V. M., Button, K. S., Chambers, C. D., Percie du Sert, N., Simonsohn, U., Wagenmakers, E.-J., Ware, J. J., & Ioannidis, J. P. A. (2017). A manifesto for reproducible science. Nature Human Behaviour, 1(1), 0021. https://doi.org/10.1038/s41562-016-0021

Newcombe, R. G. (1998). Two-sided confidence intervals for the single proportion: Comparison of seven methods. Statistics in Medicine, 17(8), 857–872.

Pager, D. (2007). The use of field experiments for studies of employment discrimination: Contributions, critiques, and directions for the future. The Annals of the American Academy of Political and Social Science, 609(1), 104–133. https://doi.org/10.1177/0002716206294796

Pedulla, D. S. (2014). The positive consequences of negative stereotypes: Race, sexual orientation, and the job application process. Social Psychology Quarterly, 77(1), 75–94. https://doi.org/10.1177/0190272513506229

Phelps, E. S. (1972). The statistical theory of racism and sexism. American Economic Review, 62(4), 659–661.

Popper, K. R. (2002). The logic of scientific discovery. Routledge. (Original work published 1959)

Porter, T. M. (1995). Trust in numbers: The pursuit of objectivity in science and public life. Princeton University Press.

Quillian, L., Pager, D., Hexel, O., & Midtbøen, A. H. (2017). Meta-analysis of field experiments shows no change in racial discrimination in hiring over time. Proceedings of the National Academy of Sciences, 114(41), 10870–10875. https://doi.org/10.1073/pnas.1706255114

Rivera, L. A. (2012). Hiring as cultural matching: The case of elite professional service firms. American Sociological Review, 77(6), 999–1022. https://doi.org/10.1177/0003122412463213